

Public Data Insights Dashboard

WEEK-04

Objective:

The aim of this project is to analyze the cleaned Netflix dataset using different data visualization techniques. These visualizations help identify trends, compare different categories, and uncover patterns within the data. Presenting the information graphically makes it easier to understand the dataset and draw meaningful insights for better decision-making. In this project, I used a Streamlit application to develop an interactive dashboard that allows users to explore the data through dynamic filters and real-time visualizations. Another tool commonly used for creating dashboards is Power BI, which provides an easy drag-and-drop interface for building interactive reports. However, Streamlit was used in this project because it offers greater flexibility by allowing dashboards to be developed entirely in Python and easily customized according to project requirements.

Dataset Overview:

The analysis was performed on the cleaned Netflix dataset containing information about movies and TV shows available on Netflix.

Data Validation

Before creating visualizations, the cleaned dataset was verified to ensure data quality.

The following checks were performed:

- Missing value inspection
- Duplicate record detection
- Column verification
- Dataset preview

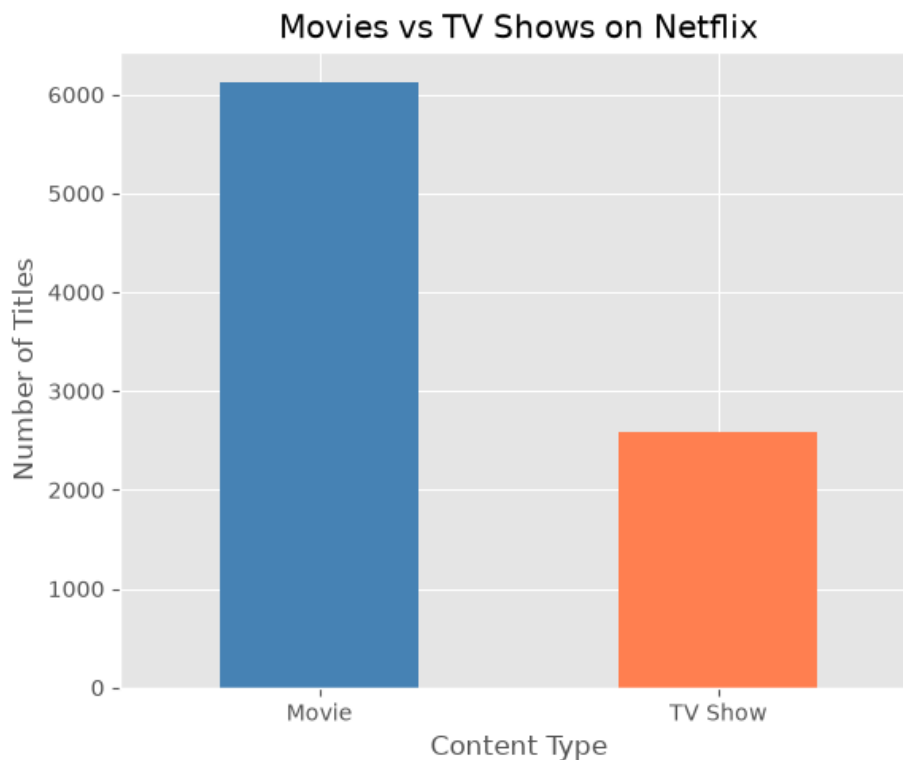
Visualizations Performed:

I developed multiple data visualizations to perform exploratory analysis on the Netflix dataset. These visualizations were designed to identify trends, compare different categories, analyze data distributions, and extract meaningful insights that support data-driven decision-making.

Bar Chart – Movies vs TV Shows

To compare the number of Movies and TV Shows available on Netflix.

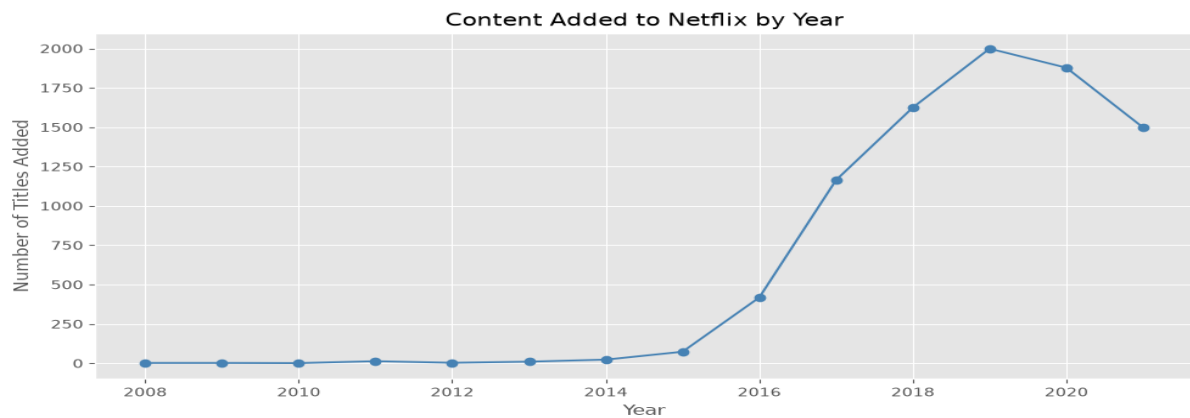
To analyze how many titles were added to Netflix each year.



The chart clearly shows that Movies dominate Netflix's catalog, accounting for approximately 70% of all available content, while TV Shows represent a much smaller portion.

Line Chart – Content Added Over Time

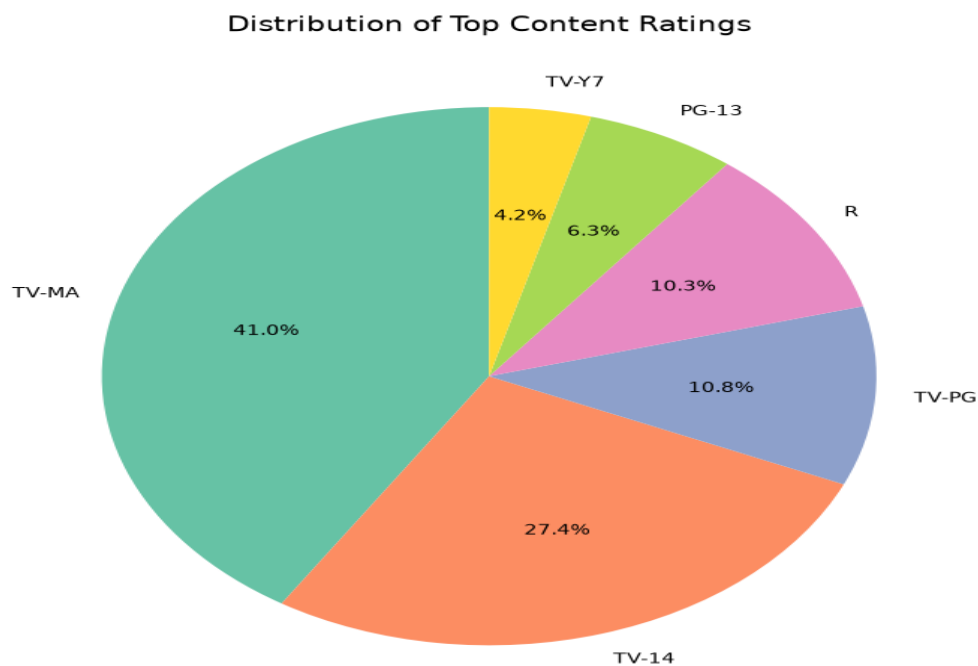
To analyze how many titles were added to Netflix each year.



The number of titles added increased steadily over the years and reached its highest point around 2019. After this peak, the number of additions declined slightly during 2020 and 2021.

Pie Chart – Top Content Ratings

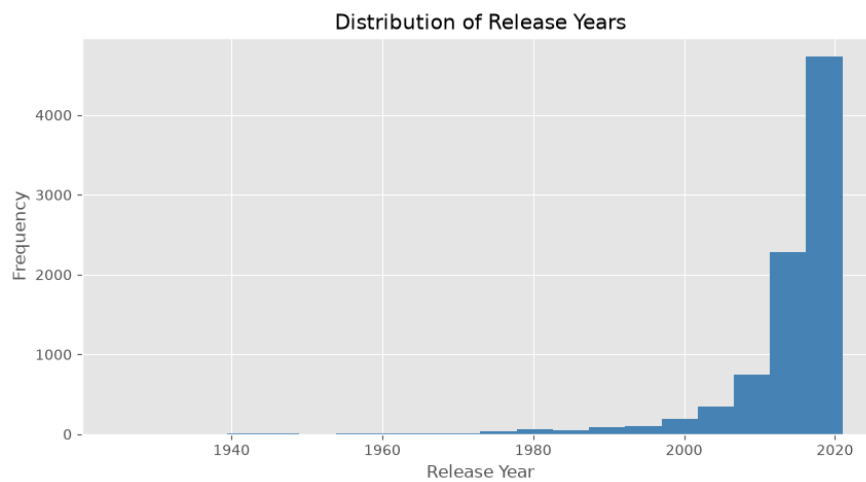
To examine the distribution of the most common content ratings.



TV-MA occupies the largest portion of the chart, followed by other ratings such as TV-14 and TV-PG.

Histogram – Release Year Distribution

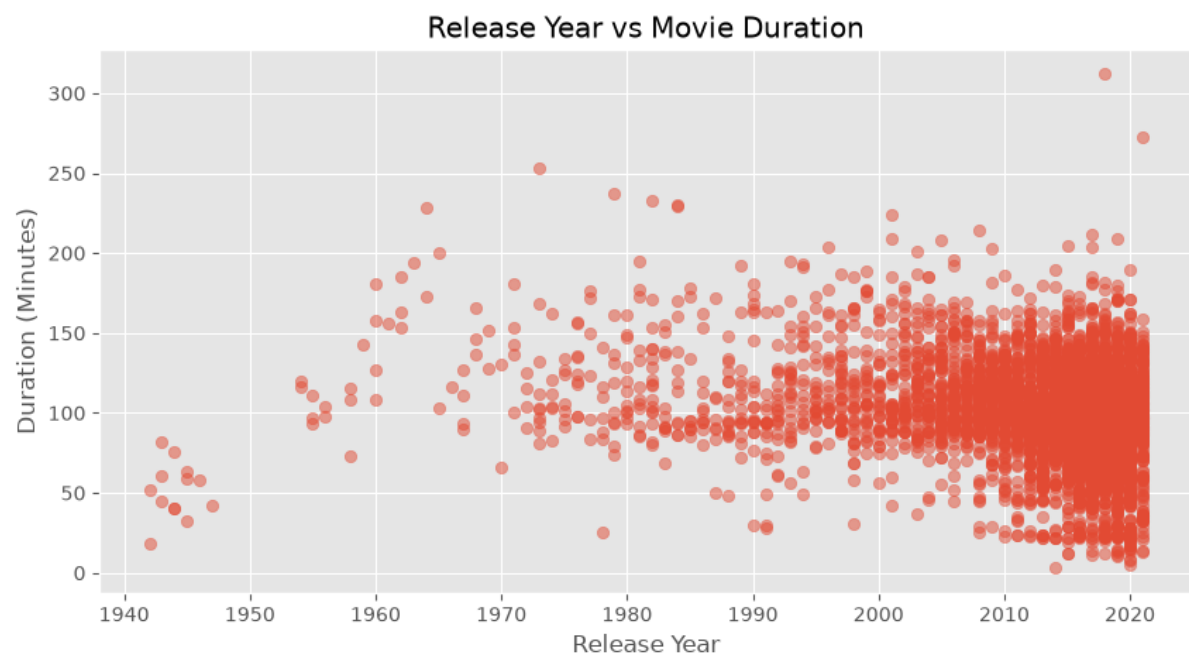
To observe how Netflix content is distributed across different release years.



Most titles were released after 2013, while comparatively fewer titles belong to earlier years.

Scatter Plot – Release Year vs Movie Duration

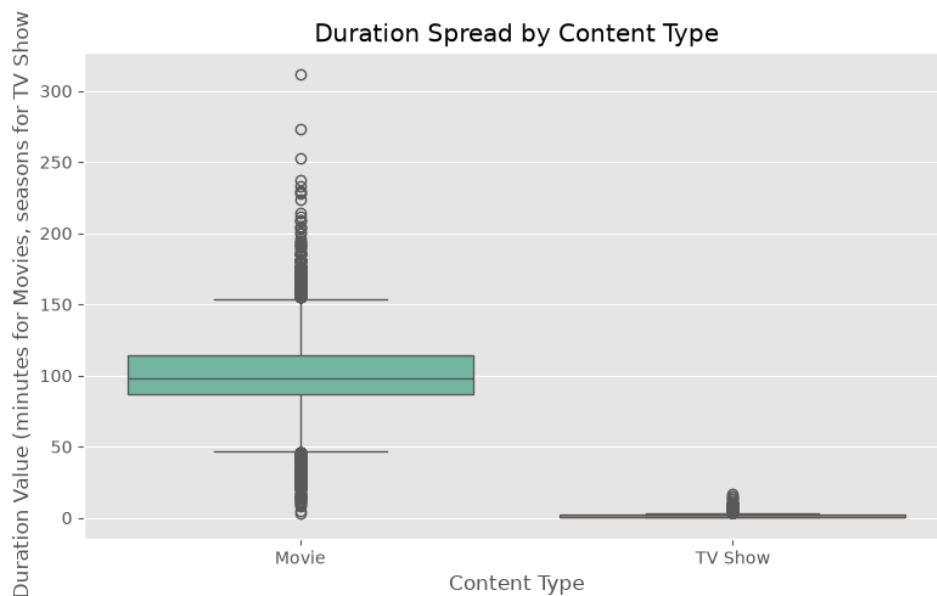
To investigate whether movie duration has changed over time.



The scatter plot shows a slight downward trend, indicating that newer movies tend to be somewhat shorter.

Box Plot – Duration Comparison

To compare the distribution of movie durations and TV show seasons.



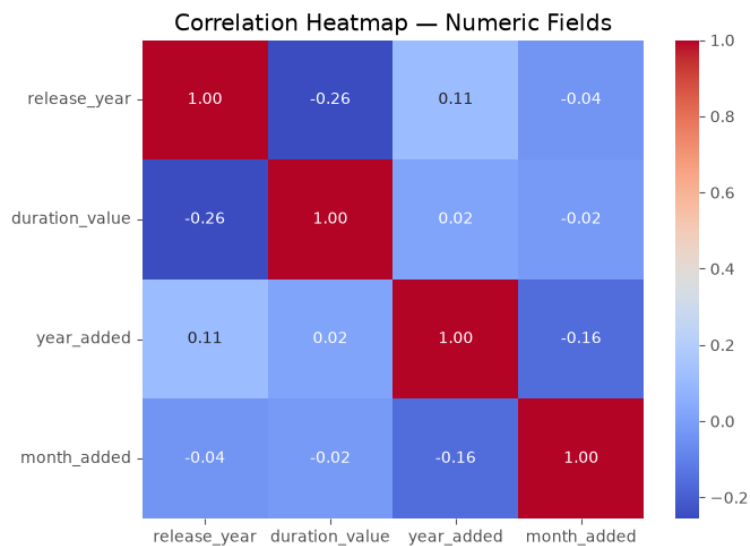
Movies generally range between approximately 87 and 114 minutes, while most TV Shows consist of only one or two seasons, with a few long-running outliers.

Heatmap – Correlation Analysis

To identify relationships among numerical variables.

Variables analyzed:

- Release Year
- Duration Value
- Year Added
- Month Added

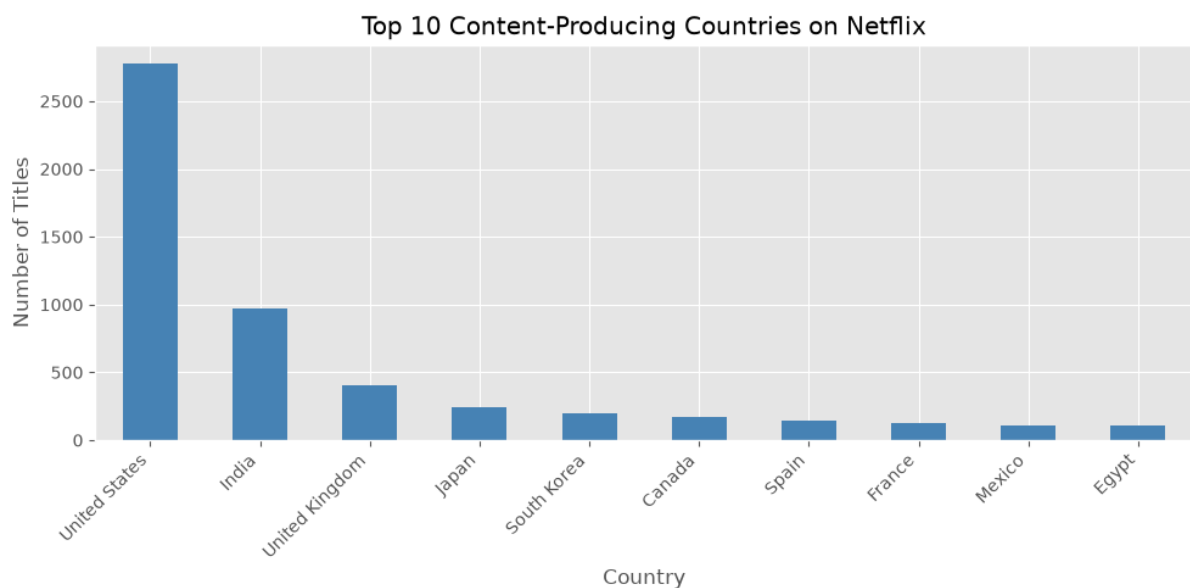


The heatmap shows mostly weak correlations between variables.

The strongest relationship appears between release year and duration, although it remains relatively weak.

Category Comparison – Top Producing Countries

To compare the countries contributing the highest number of Netflix titles.



The United States contributes the largest number of titles, followed by India and several other countries.

Streamlit-app

The Streamlit dashboard was developed to provide an interactive platform for exploring and analyzing the cleaned Netflix dataset. It allows users to

visualize the data through various charts and graphs while applying filters based on content type, release year, and country. The dashboard dynamically updates all visualizations according to the selected filters, making data exploration simple and interactive.

Dashboard Features

Interactive Filters: I added three filtering options to the dashboard to allow users to explore the dataset dynamically.:

- Content Type
- Release Year Range
- Country

Dataset Overview

I added a summary section to the dashboard that provides users with a quick overview of the filtered dataset before exploring the visualizations.. It includes:

- Total Titles
- Number of Movies
- Number of TV Shows
- Countries Represented

Data Visualizations

This dashboard contains multiple interactive visualizations that present different aspects of the Netflix dataset, including:

- Top 10 Countries by Number of Titles
- Titles Released Over Time
- Distribution of Content Ratings
- Distribution of Release Years

- Release Year vs. Movie Duration
- Duration Spread by Content Type
- Correlation Heatmap
- Top Genres

Insights

I added an Insights section to summarize the key observations obtained from the dashboard analysis. This section highlights important trends, such as the dominance of movies over TV shows, the prevalence of mature-rated content, and the significant contribution of the United States and India to Netflix's content library. It also points out areas where the platform has comparatively less content, such as family-oriented titles and content from underrepresented countries..

Recommendations

I included a Recommendations section to provide data-driven suggestions based on the analysis. The recommendations are organized into **High**, **Medium**, and **Low** priority levels and suggest expanding the TV show catalog, increasing family-friendly content, and investing in content production from underrepresented countries to improve the diversity of Netflix's content library.

Conclusion

This assignment helped me understand how data visualization can transform a cleaned dataset into meaningful insights. By creating different charts and developing an interactive Streamlit dashboard, I learned how to identify trends, compare categories, and analyze patterns in the Netflix dataset. I also gained practical experience in using Streamlit to build interactive dashboards and present data in a clear and user-friendly way.

Through this project, I improved my understanding of exploratory data analysis, dashboard development, and data storytelling.